Unbounded Wind Fields for Stratospheric Balloon Simulation

Generating realistic weather beyond a single model window

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github.com/sdean-group/balloon-env-dev

1 Why wind needs a generative world

A global balloon needs weather beyond a single training crop.

Altitude control exploits coherent differences between 18 wind levels. ERA5 is realistic but finite; BLE-VAE learns one regional patch. We need reproducible new weather anywhere a simulator asks.

D01

One seed, a world without patch boundaries

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What this work contributes

- A conditional 4D diffusion model for hourly winds on 18 height levels.
- InfiniteDiffusion adapted to coordinate-indexed wind queries.
- Seed-consistent, spatially unbounded generation without building a globe first.
- An evaluation organized around distributions, spatial physics, vertical structure, and time.

2 The 4D base diffusion model

Learned distribution: $p(x \mid \text{latitude, longitude, date, hour})$, where $x \in \mathbb{R}^{4\times 36\times 64\times 64}$.

D02

What enters the model, and what one sample contains

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Architecture

Factorized space-time U-Net: 2D residual blocks process each frame; zero-initialized 1D temporal convolutions couple the four hours. EDM trains the network to remove a sampled noise level with one joint denoising loss.

Training data and conditioning

ERA5 model levels 49–66 over the NE Pacific; 2020–2023 training weather. Clean per-pixel latitude/longitude channels locate every tile. Annual, semiannual, and diurnal harmonics condition every frame.

Does the bounded model resemble ERA5?

D03

ERA5 vs. base diffusion: same place, level, and four hours

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Use one representative held-out condition, common color scale, and visible speed maxima. Show all four hours; heights are modeled as **separate channels**, not convolved as a spatial axis.

Next for the base model Condition on a vertically resolved coarse synoptic field so large-scale regimes are supplied while diffusion learns the unresolved detail and extremes.

3 InfiniteDiffusion for wind

A bounded denoiser becomes a lazy global field by sharing noise, overlap, and cached intermediate states across local queries.

Unbounded coordinates | random access | same seed + same coordinate = same wind

One query, without the dense equation

Given: requested region R, diffusion levels $k = K, \ldots, 0$, bounded denoiser D_k, deterministic seed s, and conditioning c (coordinates + time).

1. **Cover** R with overlapping 64×64 windows Q_i and include the larger parent windows needed at level $k+1$.
2. **Reuse** every cached parent state $x_{k+1}[Q_i]$; create only missing coordinate-indexed noise from s.
3. **Denoise** each window: $y_i = D_k(x_{k+1}[Q_i], c_i)$.
4. **Blend** overlap with edge taper w_i, normalize by total weight, and cache the result $x_k[R]$.

R: query; Q_i: local window; k: noise level; c_i: local condition; w_i: edge taper; cache: previously computed state at fixed coordinates.

D04

Dependency pyramid: generate only what the query needs

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Read top-to-bottom from high noise ($k = K$) to clean wind ($k = 0$). A second query should visibly reuse the overlapping branch already in cache.

What unbounded generation should demonstrate

D05

A 64x64 denoiser answers nested queries in one virtual world

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Red boxes and diagonal corner connectors make each zoom explicit. Every view is regenerated from coordinates and the same seed, not cropped from a precomputed global canvas.

D06

Single tile vs. naive averaging vs. InfiniteDiffusion

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Comparison to read at a glance: naive independent tiles disagree in overlap and averaging blurs or reveals seams. InfiniteDiffusion coordinates the diffusion trajectory itself, so shared locations remain coherent. The same idea tiles in x and y now; extending the cache and dependency graph along t gives full spatiotemporal tiling.

Next for InfiniteDiffusion Tile four-hour blocks along time, validate continuity at temporal boundaries, and report the same multi-tile-minus-single-tile penalty used for spatial seams.

4 Evaluating realistic winds

No single score defines realism. We compare every generator with held-out ERA5 and ask which physical or statistical failure each metric exposes.

What each comparison means

ERA5 is the reanalysis target. **ERA5 floor** compares two disjoint held-out subsets and shows natural sampling variation, not a model.
Fitted simplex is unlimited procedural noise whose moments, correlations, and scale are tuned to ERA5; it cannot answer location/date-conditioned queries.
BLE-VAE is the previous learned regional generator: one bounded patch and ten levels. Extending it would require an external averaging scheme because it has no diffusion trajectory to coordinate.

Four complementary views of realism

Distributions
$W_1(u)$, $W_1(v)$, $W_1(|V|)$; 1% and 0.1% tail error; condition-matched W_1. *Are speeds and extremes plausible?*

Spatial physics
Spectral residuals of energy, divergence, and vorticity; effective resolution; shear. *Is energy placed at the right scales?*

Vertical structure
Opposing-wind fraction; jet area and elongation. *Can altitude changes expose meaningfully different winds?*

Temporal physics
Temporal spectral residual; tracer log-MSD and final spread. *Do winds persist and transport particles realistically?*

A diagnostic subset of the full evaluation

view	metric	ERA5 floor	simplex	BLE	ours
spatial	$\mathrm{SR}_E \downarrow$	0.25	0.75	2.73	**1.53**
	$\mathrm{SR}_{\mathrm{div}} \downarrow$	0.25	0.87	5.23	**1.56**
distribution	$W_1(\text{condition}) \downarrow$	2.38	3.38	N/A	**3.52**
	0.1% tail error ↓	2.92	2.34	18.94	**13.54**
vertical	opposing-wind error ↓	0.00	0.02	0.37	**0.16**
	jet elongation $\to 1$	1.17	0.85	1.76	**0.95**

How to read it: closer to the ERA5 floor is better. Different rankings are useful: our model improves on BLE spatially and in jet shape, while fitted simplex still wins spectra, tails, and opposing winds. Conditioning has not yet closed the distribution gap. Temporal rows await long-horizon tiling.

D07

From a power spectrum to one spectral-residual score

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Extent is solved; realism is still limited by the base denoiser.

Current evidence: seamless spatial queries do not damage the metric suite, but the base model is too smooth, under-produces extremes, and rarely creates opposing winds across height.

Next for evaluation Use a simplified baseline set in every plot; add long-horizon temporal metrics after time tiling; evaluate whether the resulting wind worlds improve balloon policies.

Goslin et al., *InfiniteDiffusion* (NeurIPS 2022) · Karras et al., EDM · Hersbach et al., ERA5 (QJRMS 2020)